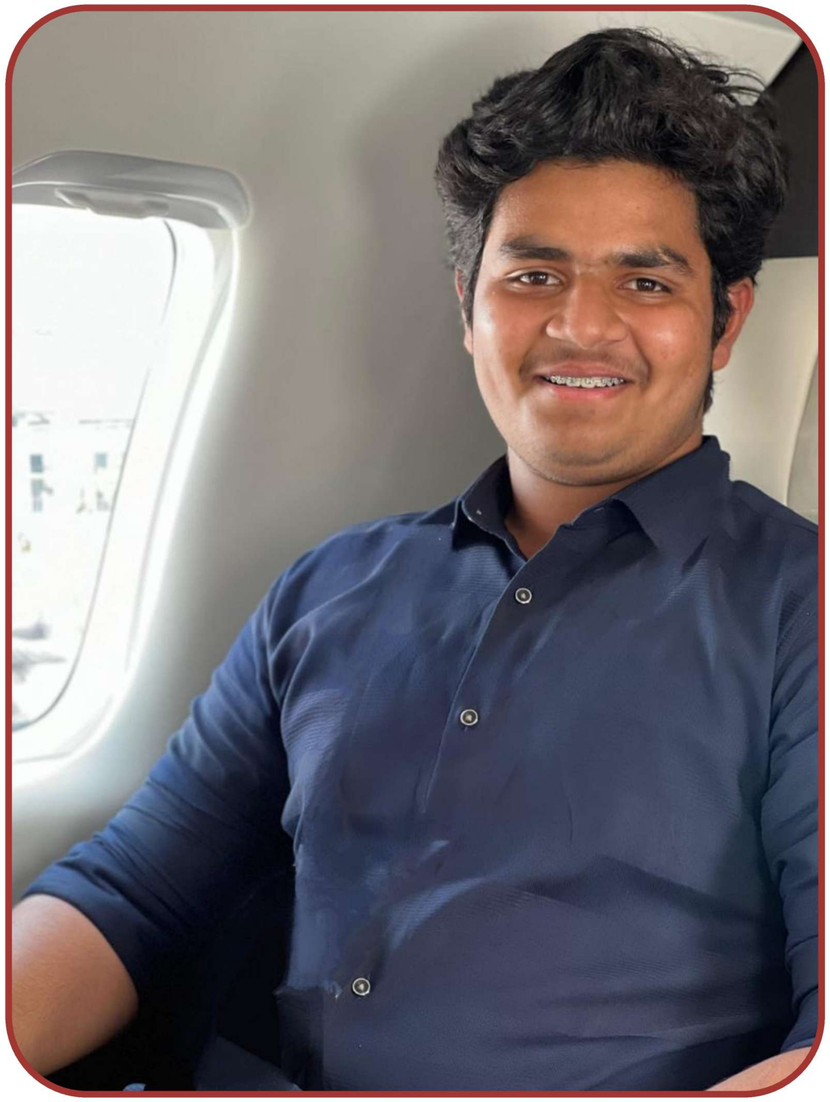
portfolio



Naga Adithya

Architecure Student

Hello.
I’m Adithya!

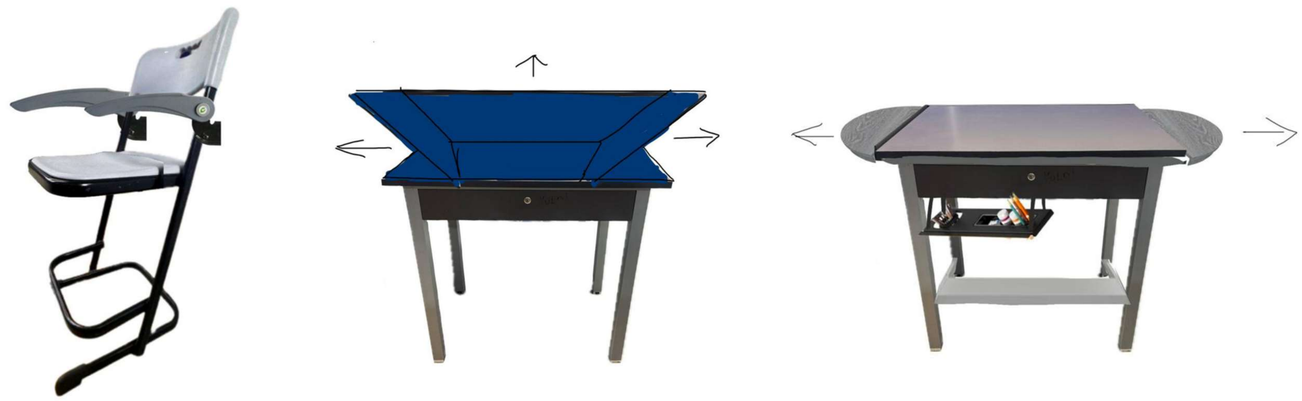


I am an architecture student interested in how design can respond to real challenges in everyday environments. My approach focuses on understanding problems, observing how people interact with spaces, and developing practical solutions that improve comfort, safety, and usability.
Across my projects, I aim to create designs that are adaptable, efficient, and responsive to their context. Whether working on small scale objects or larger environments, I prioritise user experience, functionality, and long term value while exploring solutions that support people and their everyday activities.

Upcycling Furniture

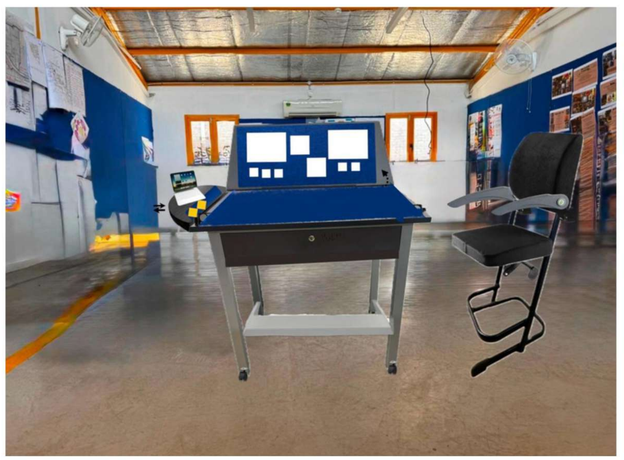
For this project, I redesigned the classroom workstation by focusing on the everyday challenges architecture students face in the studio. The existing tables were difficult to move, lacked space for tools when the drafting surface was lifted, and did not provide enough pin up area for presenting work.
My goal was to create a workstation that supports the different ways students use their desks throughout the day. The final design includes controlled wheels for quiet movement, a foldable side table for tool storage during drafting, and an integrated foldable pin board within the tabletop. This creates a flexible workspace that supports drafting, storage, and presentation in a single unit.

The process began by studying different workstation designs used in studios and workspaces. I analysed how they function and created mise en scènes to compare their features, movements, and spatial organisation.



From this analysis, I identified useful elements that could improve flexibility and usability. I then adapted these ideas to suit the specific needs and layout of our classroom so that the design would integrate with the existing studio environment.





Upcycling Furniture

The final workstation was designed to support the different activities that take place during studio, including drafting, reviewing drawings, storing tools, and presenting work. Each element of the design responds directly to problems identified during observation of everyday studio use.

Tilt Activated Mobility

Caster wheels are placed at the front of the table and activate only when the desk is tilted forward, allowing quiet and controlled movement within the classroom.

Foldable Side Surface

A foldable side table provides additional space for tools and materials while drafting, helping keep the main work surface clear.

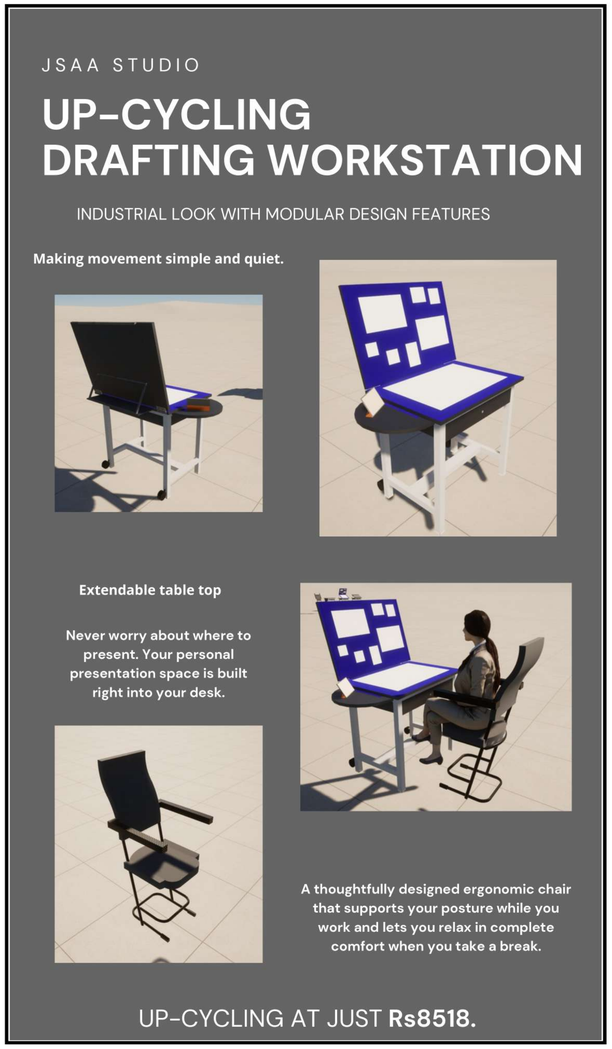
Integrated Pin Board

A foldable pin board is built into the tabletop, allowing students to quickly display drawings and work during discussions or reviews.

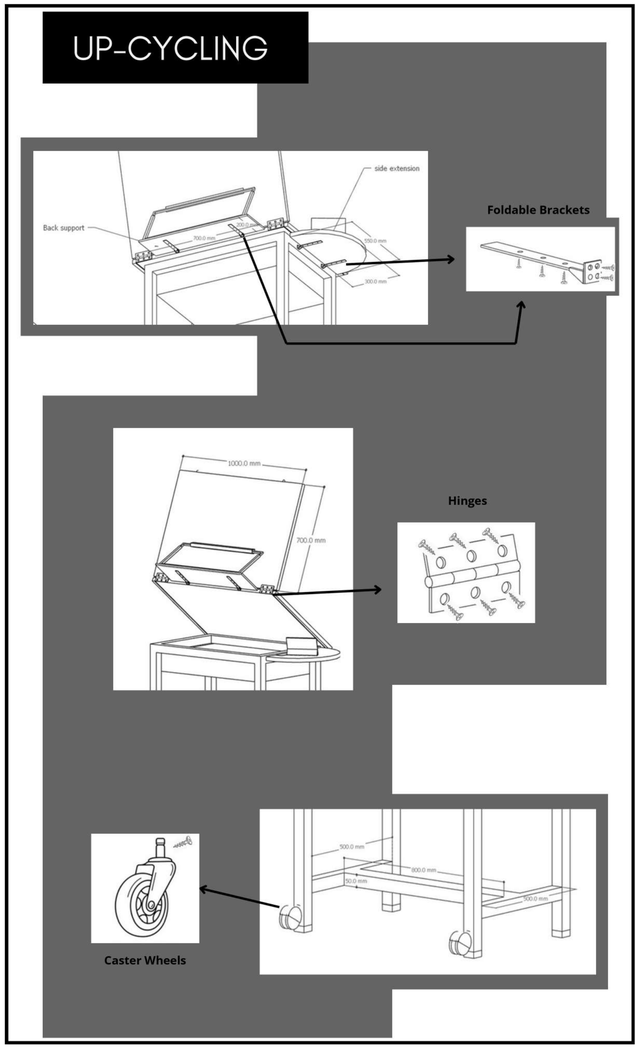
Multi Functional Workstation

The design supports drafting, storage, collaboration, and presentation within a single adaptable desk.

Sales Poster



DIY Poster





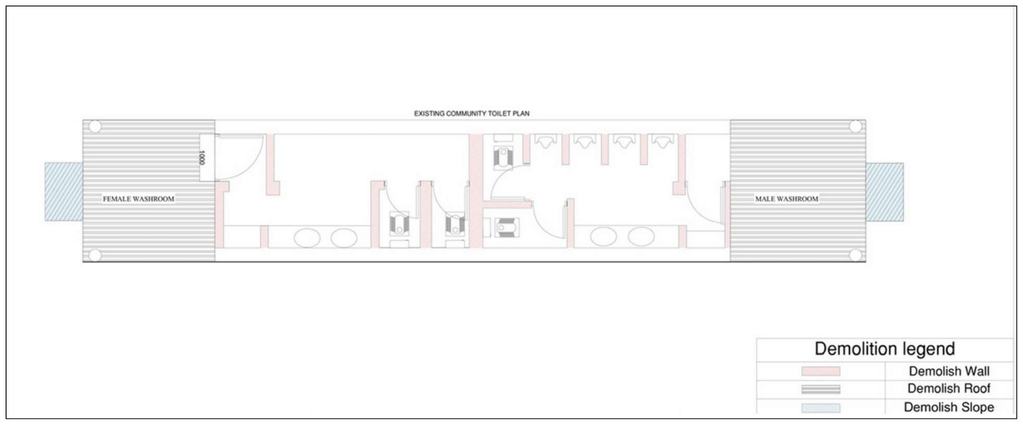
This image illustrates how the workstation functions and how its different components are used during studio activities.

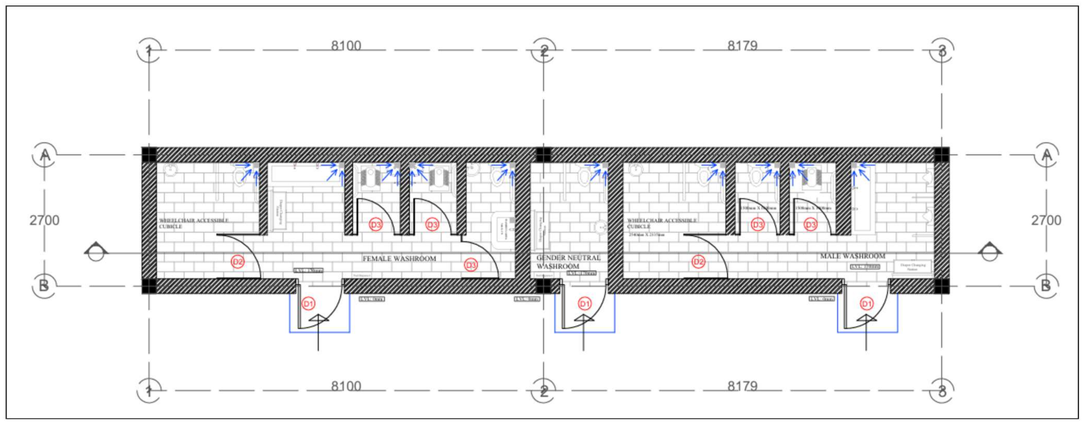
Menstrual Equity Toilet

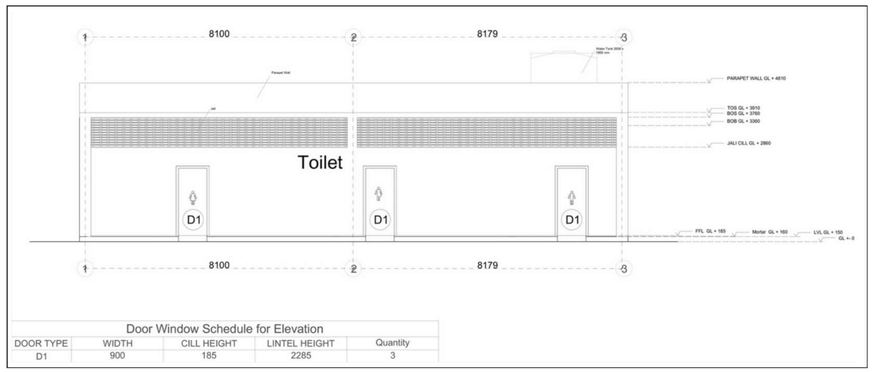
This project reimagines the public toilet as a space of dignity, safety, and accessibility by placing menstrual equity at the center of the design. Through the study of user behaviour and spatial experience, the proposal addresses issues of privacy, comfort, and inclusivity while developing a buildable prototype for a site in Sonipat.

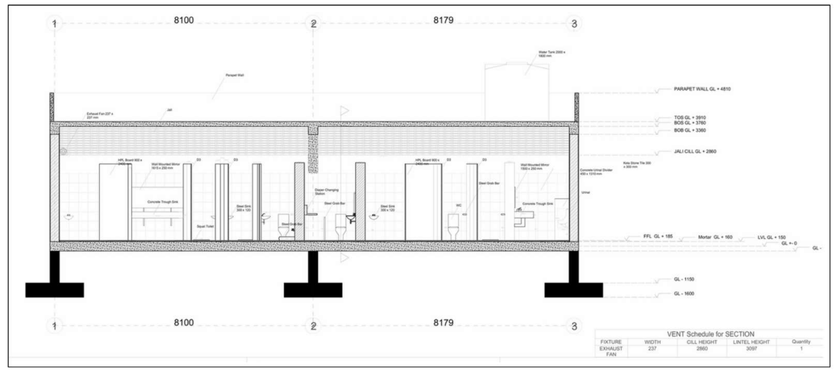
Floor Plans

In this project, we redesigned a public toilet that had major issues, especially the lack of privacy and menstrual comfort. The original entrance opened into an open ground, making users feel exposed and unsafe. We relocated the entrance to a more secluded side and reorganised the layout to create safer and more comfortable circulation.







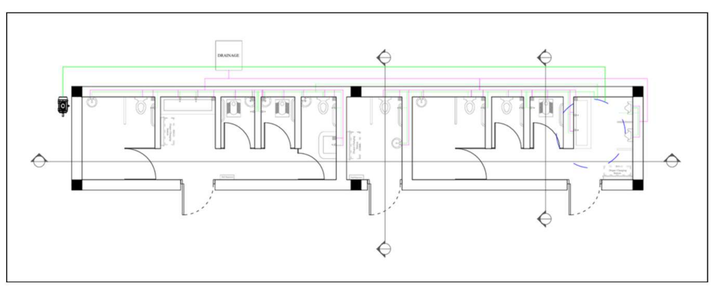


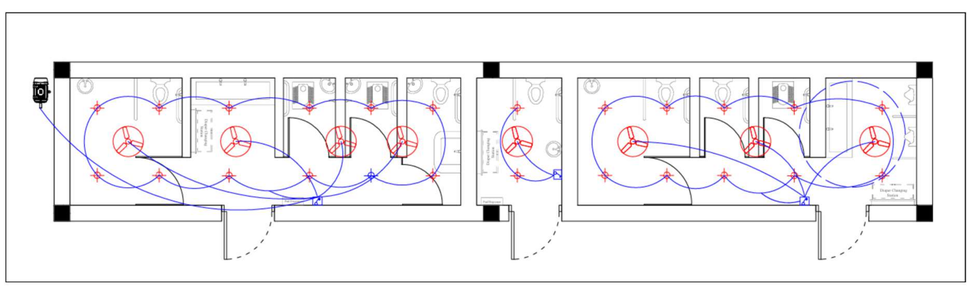
Section & Elevation

On the northern elevation, we've added a brick jali to bring in more natural light into the cubicles while maintaining privacy and reducing dependence on artificial lighting, which also lowers the maintenance cost, as this is a public toilet maintained by the government.

Service Plans

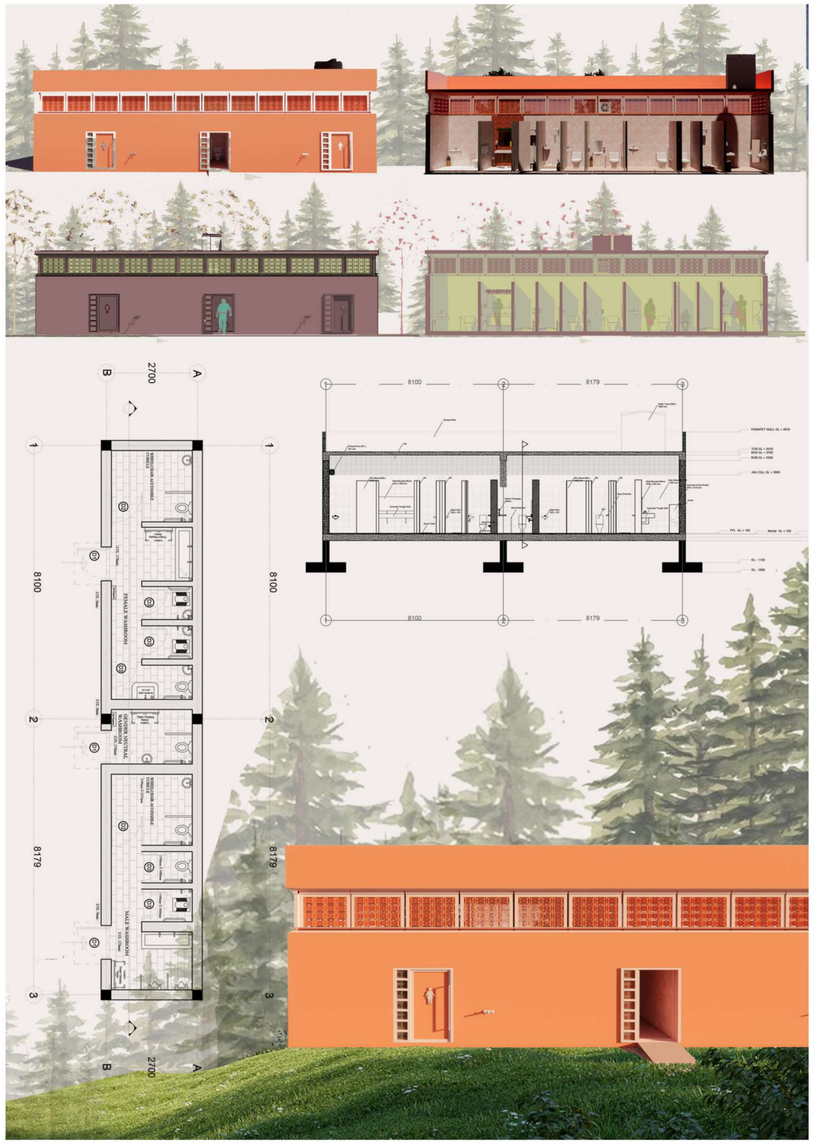
We upgraded the cubicles to be menstrual-equipped following the NBC guidelines. This redesign has transformed the space into one that is not only more dignified, safe, and user-friendly for everyone, but also specifically caters to the needs and comfort of menstruators, ensuring a respectful and inclusive environment for all users.





Menstrual Equity Toilet



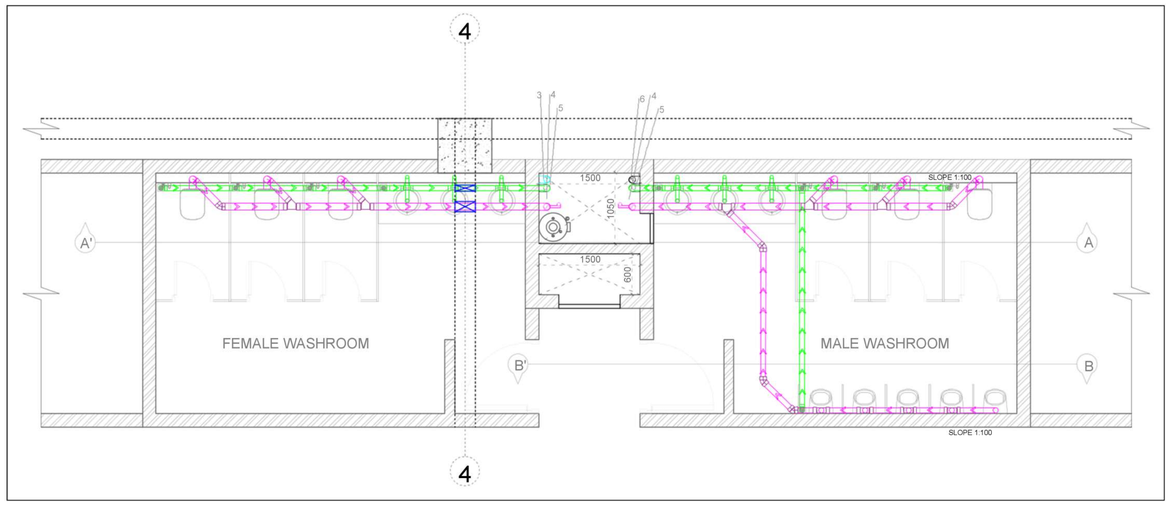


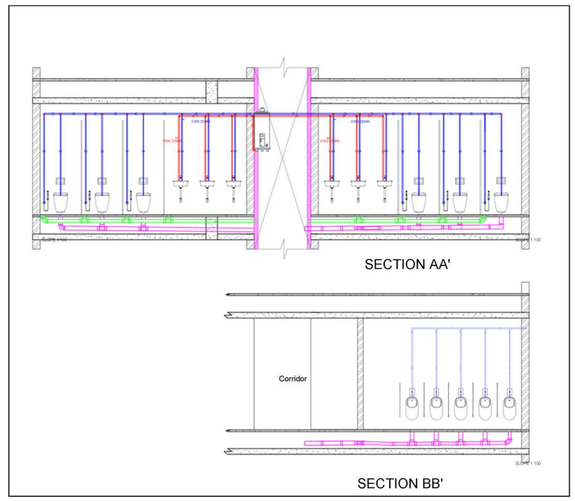
These posters present the architectural proposal of the Community Toilet through visualisations and technical drawings. They illustrate the spatial layout, elevations, sections, and overall form of the building, helping communicate how the design would appear and function in a real context.

Services

Plumbing

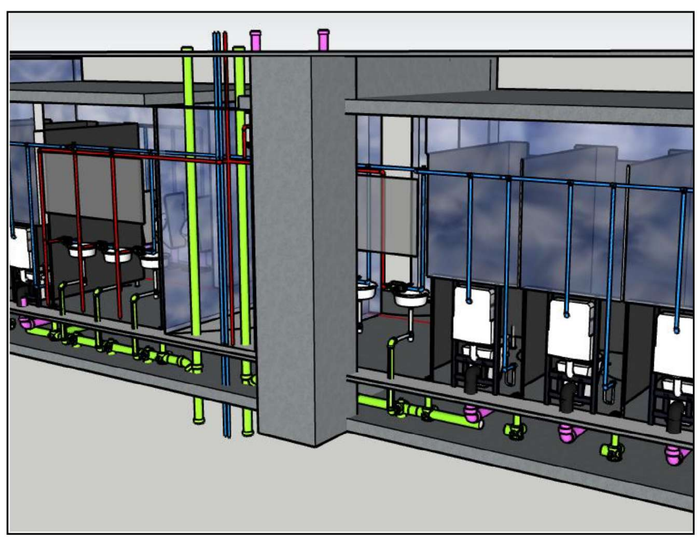
In this project, I developed a coordinated building services plan for an office building in Noida. The design integrates plumbing, fire fighting, HVAC, and electrical systems to support efficient building performance while maintaining comfort, safety, and optimal use of space.





A beam running below the slab blocked the direct route of the drainage pipe to the service shaft, creating a structural constraint that required an alternative routing solution.

To resolve this, I introduced sleeves in the beam to allow the pipe to pass through without affecting the structure, which helped me better understand drainage slopes and the coordination between structure and services.



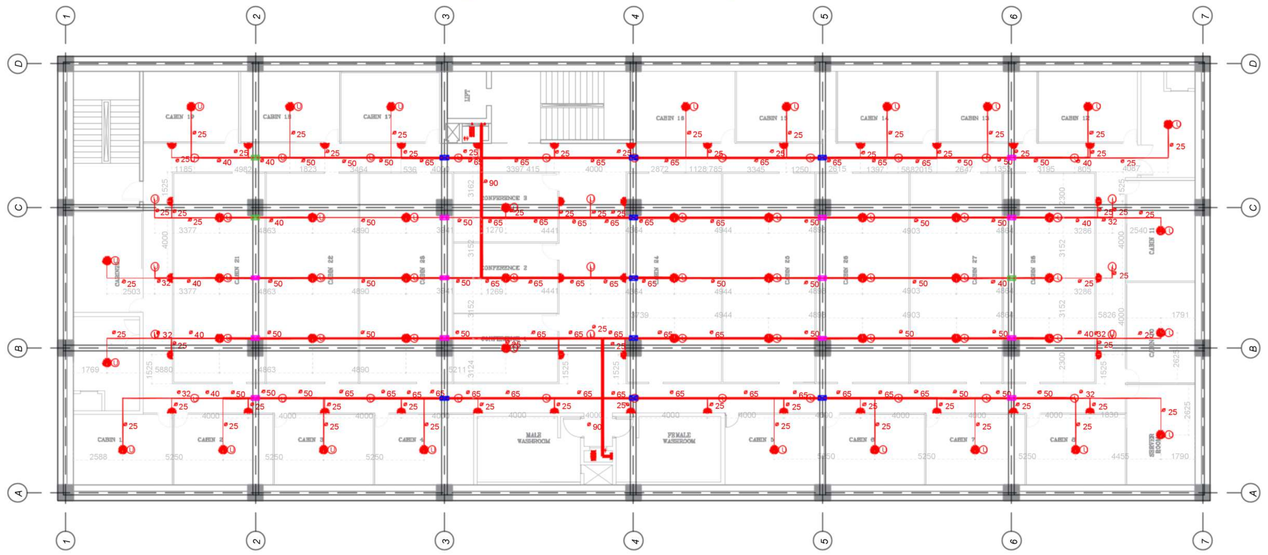




Services

Fire Fighting

Sprinkler Layout

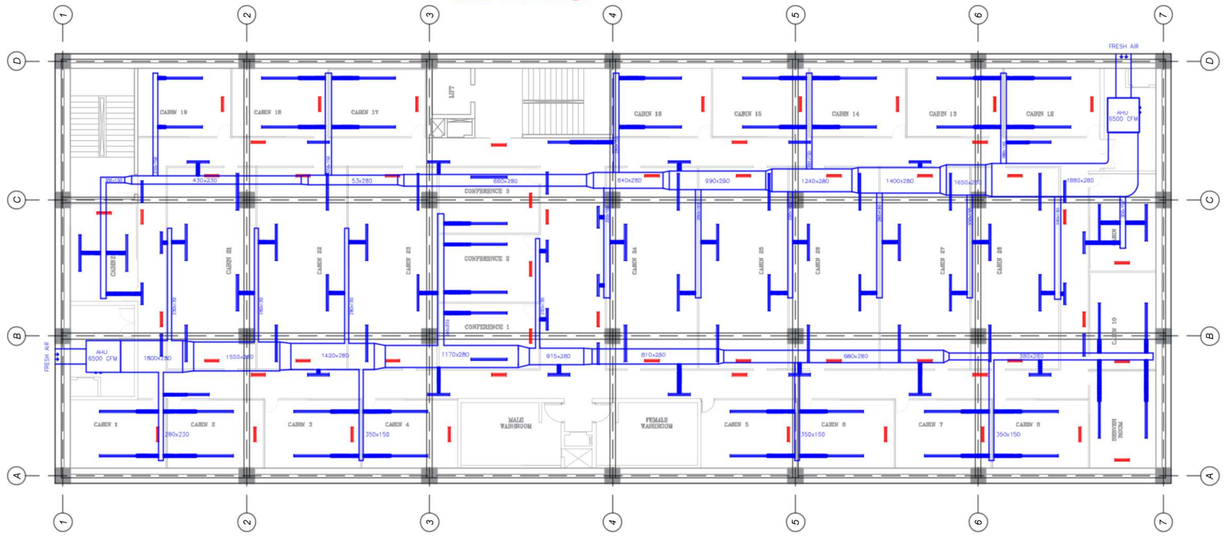


Since the office layout had very few internal walls, the fire fighting system required careful planning to ensure full coverage. This led me to study how wall mounted sprinklers work, their spray angles, and how spacing is determined. Through this process, I gained a stronger understanding of fire safety standards, the organisation of fire shafts, and the appropriate selection and placement of fire extinguishers.

BOQ

S.No	Description	Unit of Measurement	Quantity	Price	Total Amount
1	**G.I Pipes** *Supplying, installing, testing and commissioning of G.I Pipes confirming to IS IS:1239 PART-1 (2004) Heavy grade with painting, suitable type of supports,anchor fasteners,bolt nuts,clamps"U"bolts,malleable specials such as Reducers, Tees,elbows,flanges,including cutting,welding,fixing in/on walls,ceilings,by using suitable supports etc.,as per drawings.The quoted rate shall also include for chasing,chipping walls,making bore holes in walls/floor and making them good iwith filler material and finished in cement morter etc.(Make-Jindal)*				
	90mm	m	14	700	**9800**
	65mm	m	99	240	**23760**
	50mm	m	106	200	**21200**
	40mm	m	26	160	**4160**
	25mm	m	94	112	**10528**
2	**Smoke Detectors**	Each	54	1270	**68580**
3	**Pendant Sprinklers** *Supply, install, connect and test of 15 mm brass Pendant sprinkler heads to be installed at all areas above false ceiling. all as specified, as shown on drawings and as per QCDD requirements. The work shall be according NFPA 13 (Make - Safe Pro Fire Services Pvt. Ltd.)(300 MM)*	Each	49	195	**9555**
4	**Upright Sprinklers** *Supply, install, connect and test of 15 mm brass Pendant sprinkler heads to be installed at all areas above false ceiling. all as specified, as shown on drawings and as per QCDD requirements. The work shall be according NFPA 13 (Make - Safe Pro Fire Services Pvt. Ltd.)(300 MM)*	Each	80	440	**35200**
5	**Side-wall Sprinklers** *Supply, install, connect and test of 15 mm brass Pendant sprinkler heads to be installed at all areas above false ceiling. all as specified, as shown on drawings and as per QCDD requirements. The work shall be according NFPA 13 (Make - Safe Pro Fire Services Pvt. Ltd.)(300 MM)*	Each	38	395	**15010**
6	**Hose Pipe** *Supply,lift into position, install, test, set to work Premier extra/flyfyter super type A with EPDM rubber lining Supply,lift into position, install, test, set to work, and commission All necessary flushing connection assembly (Make- Fire Zone Pvt. Ltd.)*	Each	2	6170	**12340**
7	**Hose drum reel** *Supply,Lift into position, install, test, set to work, and commission All necessary flushing connection assembly (Make- Safe Pro Fire Services Pvt. Ltd.)*	Each	2	8000	**16000**
8	**Hose Cabinet shutter** *Supply, lift into position, install, test, set to work, and commission Fire Hose cabinet with 20 meters 25 mm (1 inch) diameter fire rubber hose cabinet assembly complete with 40 mm (1½ inch) landing valve, bronze adjustable spray nozzle, (multi-purpose dry chemical powder fire extinguisher 10-Kg capacity) (Make- Safe Pro Fire Services Pvt. Ltd.)*	Each	2	6700	**13400**
9	**Manual call point** *Supply, Erection, Testing and Commissioning of Addressable manual call station with necessary break glass hammer with chain and push button. The manual call station shall be of wall mounted with necessary mounting and fixing arrangement and etc., complete. As per IS 2189. (Make- Safe Pro Fire Services Pvt. Ltd.)*	Each	3	3700	**11100**
10	**Fire alarm Control Panel** *Supply, Erection, Testing and Commissioning of 1 Microprocessor based addressable fire alarm control panel with built in loop cards. FACP shall have capacity to lock upto minimum of 198 devices per loop i.e. 99 detectors & 99 modules, further shall be designed with 25% spare capacity towards future expansion FACP shall have 2 x 40 character LCD alphanumeric display with back light, fully (100%) programmable disablement. Test, sounder fault, delayed mode, relays disabled, earth fault, system/CPU fault. sounder disabled. Alarm silenced, supply fault, power etc.. It shall be provided with required capacity of maintainable free sealed acid batteries to provide 48 hours (including 30 mins in alarm condition) of indications with a maximum of 200 software zones. make-(Fire Zone Pvt.Ltd)*	Each	1	9550	**9550**
11	**Monitor Module** *Supply, mounting, Testing and Commissioning of Analogue Control module for integrated to master panel shall be capable of providing provision for zone panel (Make- Safe Pro Fire Services Pvt. Ltd.)*	Each	3	2400	**7200**
12	**Fire Pumps** *Split case horizontal type pumps as specified. mounted on a steel base; complete with suction and delivery manifolds, OS&Y valves, non-return. valves, strainers, gauges, flexible connections and factory wired control panel; connections to cold water suction including puddle flanges, vortex steel plates, etc. and fire protection delivery pipework, pumps test line with flow meters and gate valves, and electrical connections to local outlets/control panel; complete as shown on drawings and as specified.*				
13	**Diesel pump** *Supply, install, connect, test and commissioning an automatic, factory assembled wired and tested Diesel driven fire pumps system consisting of the pump, motor, diesel tank, and controller, (pump & motor) to be mounted on a common base prior to shipment. Fire pumps shall be horizontal split case bronze fitted, double suction, centrifugal The pump complete package shall be conforming to all requirements of NFPA 20 and as shown on drawings. (Make- Safe Pro Fire Services Pvt. Ltd.)*	Each	1	489500	**489500**
14	**Jockey pump** *Supply, install, connect, test and commissioning an automatic packaged End Suction jockey pump, complete including but not limited to the UL listed controller and all other accessories as specified and as shown on drawings. (Make- Fire Zone Pvt. Ltd.)*	Each	1	102000	**102000**
15	**Hydrant pump** *Supply, install, connect, test and commissioning an automatic packaged Hydrant pump, complete including but not limited to the UL listed controller and all other accessories as specified and as shown on drawings. (Make- Fire Zone Pvt. Ltd.)*	Each	1	210000	**210000**
16	**Sprinkler pump** *Supply, install, connect, test and commissioning an automatic packaged Sprinkler pump, complete including but not limited to the UL listed controller and all other accessories as specified and as shown on drawings. (Make- Fire Zone Pvt. Ltd.)*	Each	1	210000	**210000**
17	**Fire Extinguishers - Grade ABC - 6 kg** *Supply, lift into position, install, test, set to work, and commission fire extinguisher with the following capacity and as per drawing (Make-Kanex)*	Each	6	5810	**34860**
18	**Fire Exit - single door** *Supply, install, test of Fire exist single door with pushbar made up of Galvanised iron sheets provided along with the required fixtures for installation (Conform to IS 277) (Make- Safe Pro Fire Services Pvt.Ltd*	Each	1	18400	**18400**
	Total				**1332140**

HVAC



The HVAC system was designed to ensure efficient air distribution and thermal comfort within the office spaces. The Air Handling Unit (AHU) was strategically positioned to optimise duct routing and reduce service conflicts. Different types of diffusers and vents were used to distribute conditioned air evenly across the open plan layout. The system was developed based on load calculations and coordinated with the overall services layout to maintain efficiency and spatial organisation.

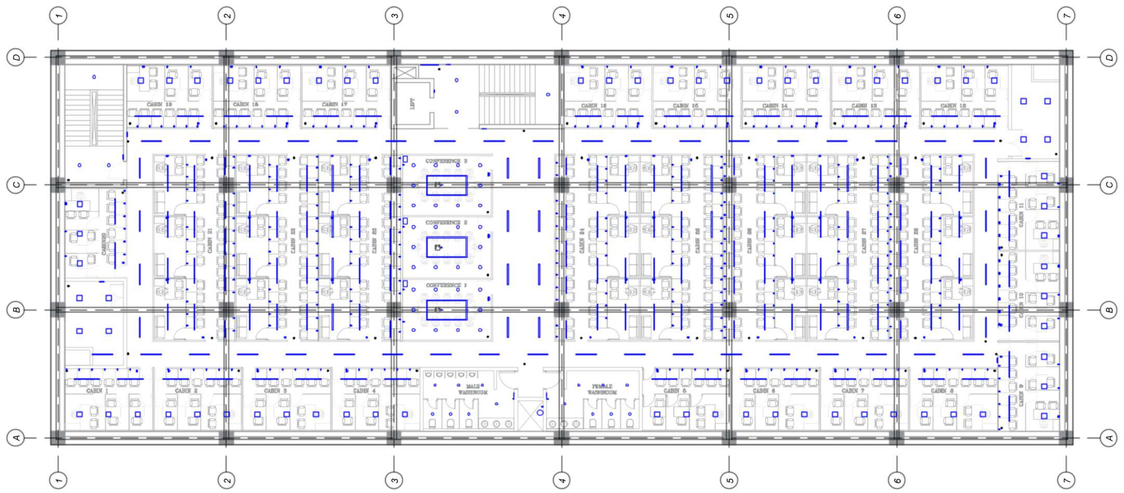
BOQ

S.No	Description	Unit of Measurement	Quantity	Price	Total Amount
1	**Air Handling Units** *Supply,Loading,Unloading, lifting-shifting, positioning, levelling & installation in place of double Skin Ceiling Mounted Air Handling Units with 25 mm Thick Sandwiched PUF insulation as per specifications of drawings.Each complete with curved centrifugal fan, filter water coil of copper tube& aluminium fins construction with blue hydraulilic coating for chilled water coil. All AHUs shall be delivered at site and installed in place by skilled workmen of AHU supplier/Contractor. (Make - Citizen by Daikin/Auracool Systems Pvt Ltd/Equivilant)*	Each	2	165000	**330000**
2	**Factory Fabricated ductwork** *Supply, fabrication, installation and testing of sheet metal ducts (factory fabricated ducts) in accordance with the approved drawings complete with all accessories like vanes, flanges,anchor bolts,GI bolts & nuts,connections, duct dampers,etc as required by the specifications, and shall also confirm to the SMACNA specifications.(Make- Auracool Systems Pvt Ltd/ Samco flex Ductworks/Equivilant)*	Sqft	1800	1000	**1800000**
3	**Slot diffusers (2000mm)** *Supply, installation and testing of powder coated aliminium Slot diffusers as per approved drawing & specifications with removable core. (Auracooling Pvt Ltd/ Air Master/ Equivilant).*	Each	78	2000	**156000**
3	**Slot diffusers (1200mm)** *Supply, installation and testing of powder coated aliminium Slot diffusers as per approved drawing & specifications with removable core. (Auracooling Pvt Ltd/ Air Master/ Equivilant).*	Each	30	1200	**36000**
4	**Return Air Diffusers/Exhausts** *Supply, installation and testing of powder coated aliminium Square return air diffusers/ Exhausts as per approved drawing & specifications with removable core.(Auracooling Pvt Ltd/ Air Master/ Equivilant).*	Each	53	800	**42400**
	TOTAL				**2364400**

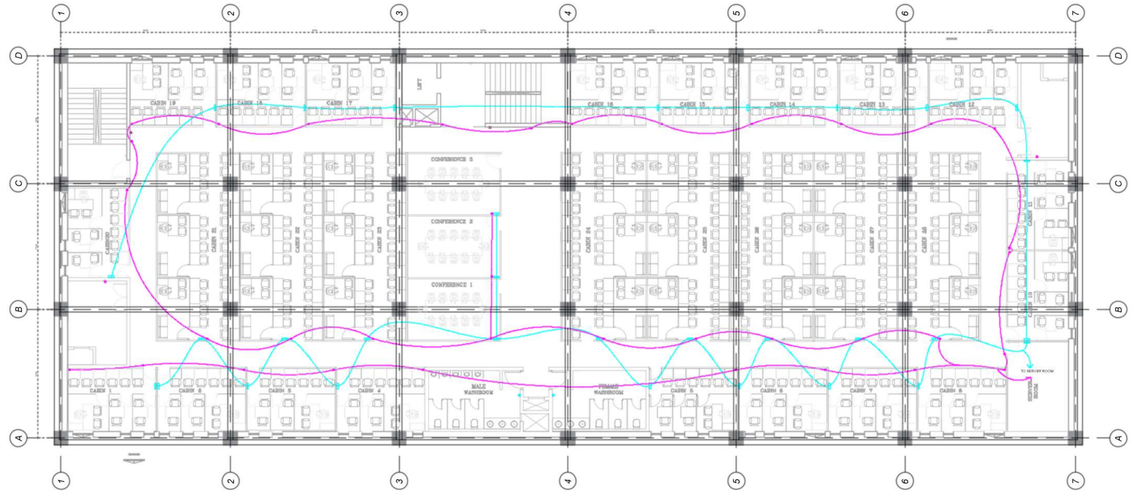
Services

Electrical

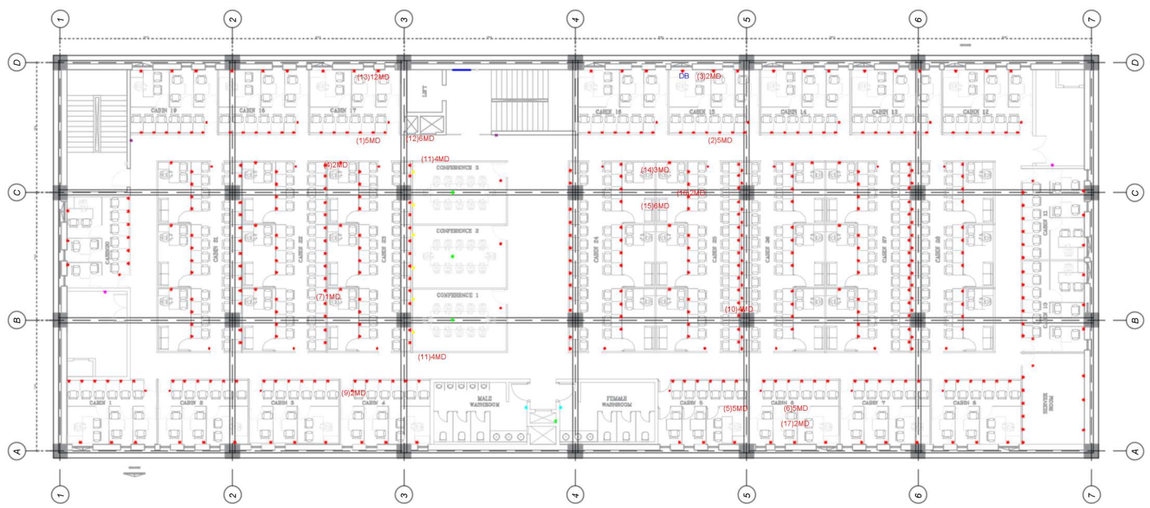
The electrical system was designed to support the functional needs of an office environment, including lighting, power supply, WiFi, CCTV, and network connections. Since the office layout had minimal internal walls, lighting fixtures were connected through the ceiling to maintain a clean and organised system. Rope pull switches were introduced where wall mounted switches were not feasible due to glass partitions. Floor mounted switchboards were also provided near workstations to ensure accessible power supply while maintaining flexibility in the open plan layout.



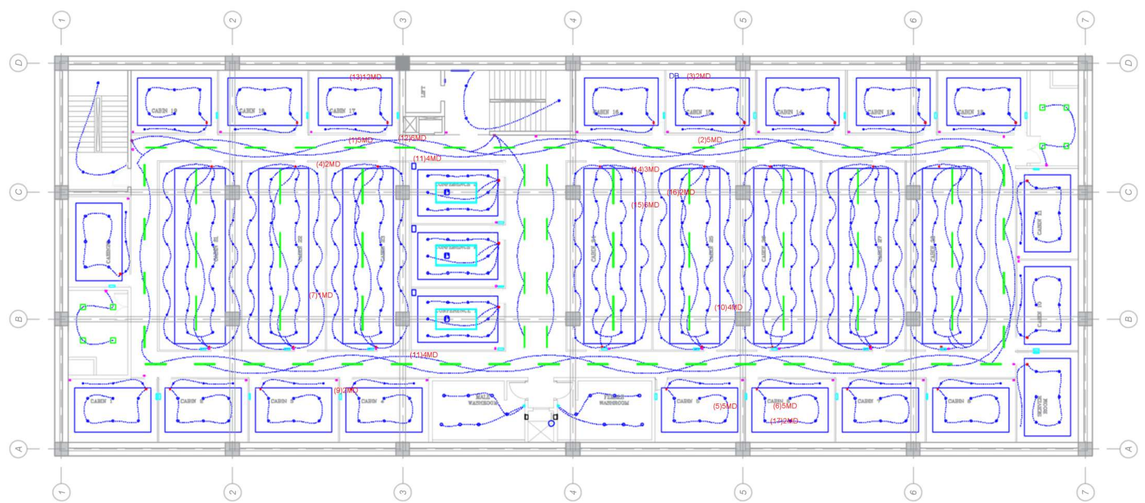
Fixture Layout



CCTV, WIFI Looping



Socket Layout



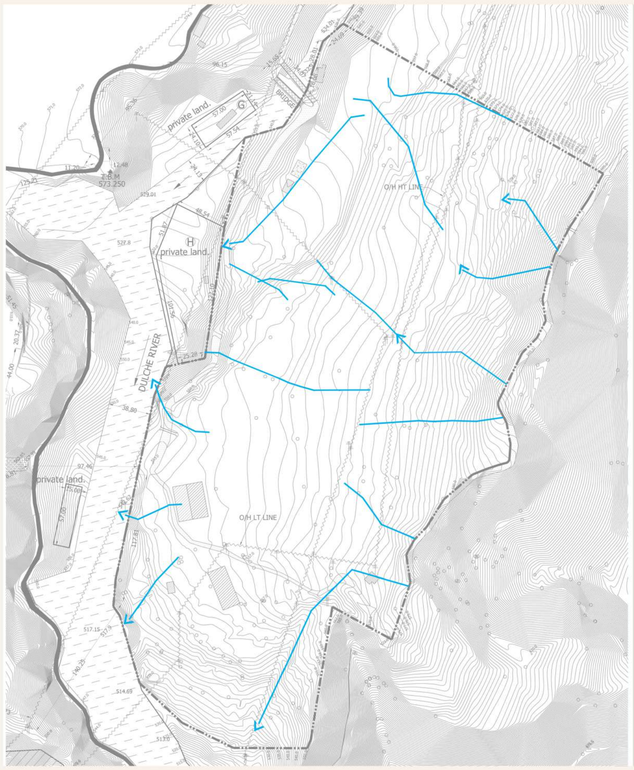
Fixture Looping

VISTAVANA
RESIDENCES

To create a sustainable, culturally rooted, and community-oriented residential environment in Mandi that enhances quality of life through thoughtful planning, local identity, and ecological sensitivity.

TOPOGRAPHICAL ANALYSIS

DRAINAGE

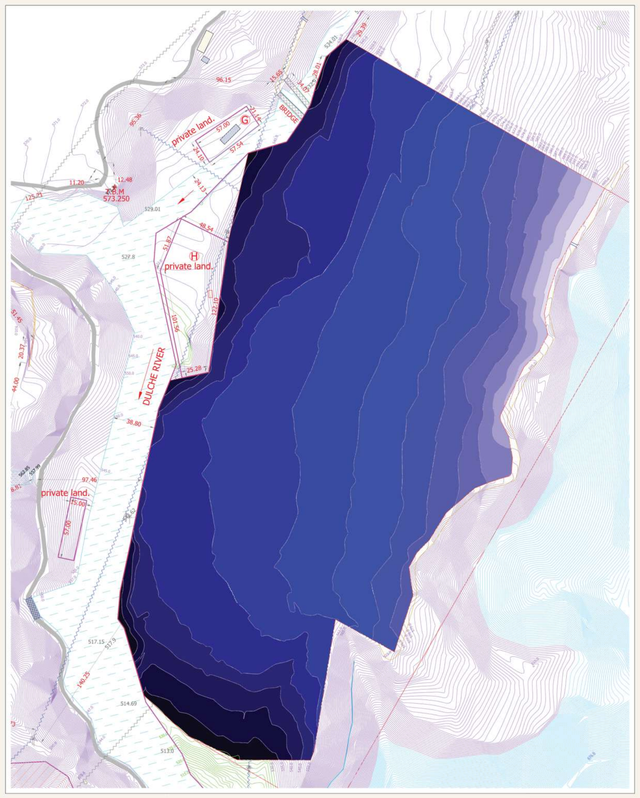


The drainage map traces natural water flow channels across the site, showing how runoff moves through the terrain.
Water-logging generally forms in the low-lying pockets between or away from the main drainage channels, where water cannot easily move toward the natural flow lines.

WATER ACCUMULATION MAPPING



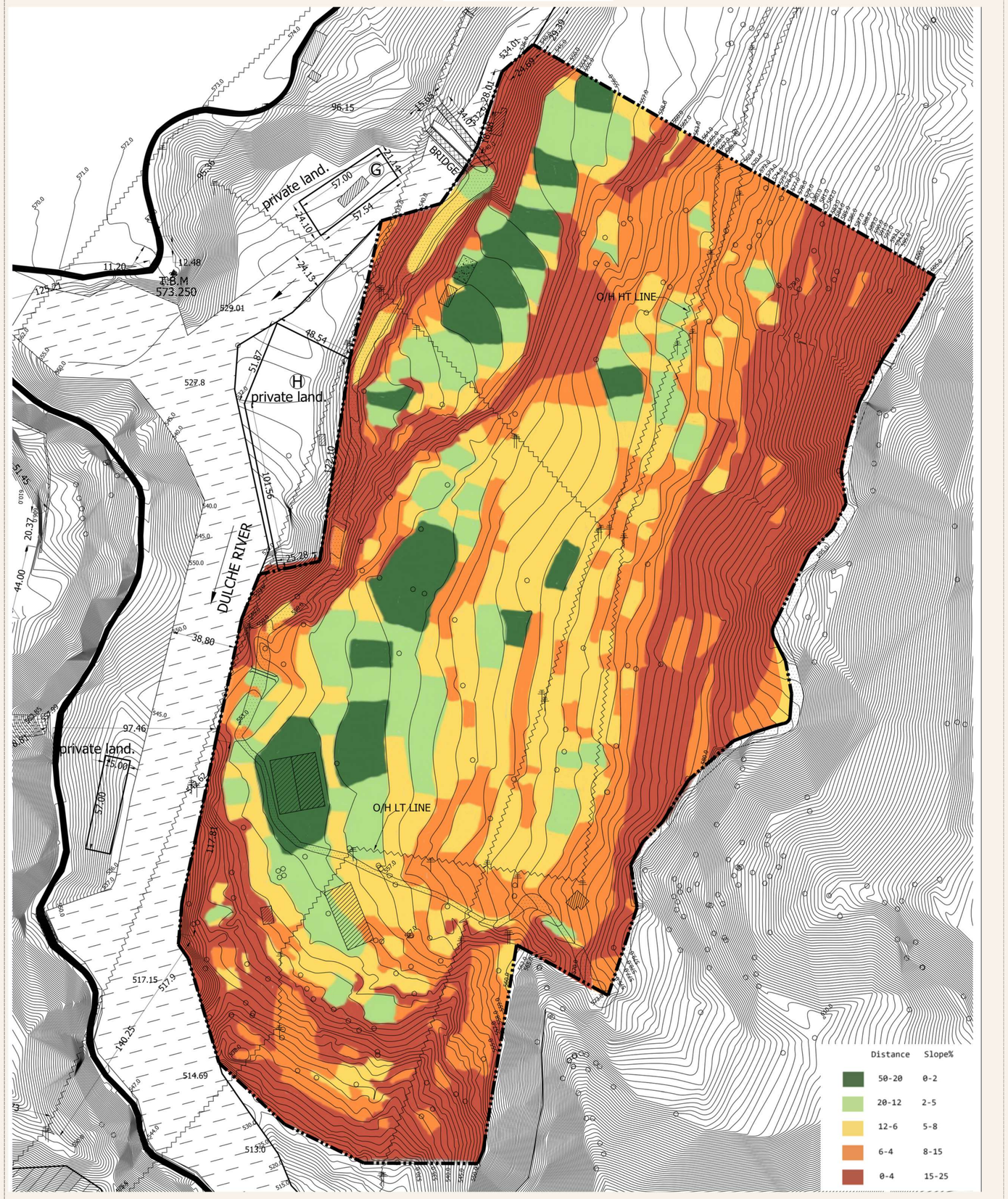
ELEVATION



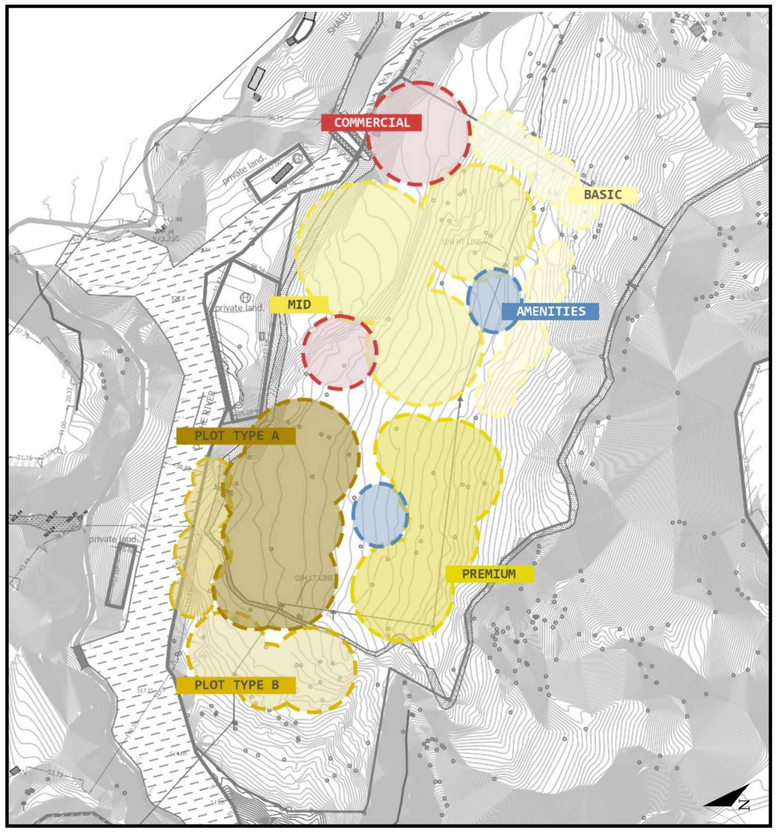
SITE ENTRY EXIT

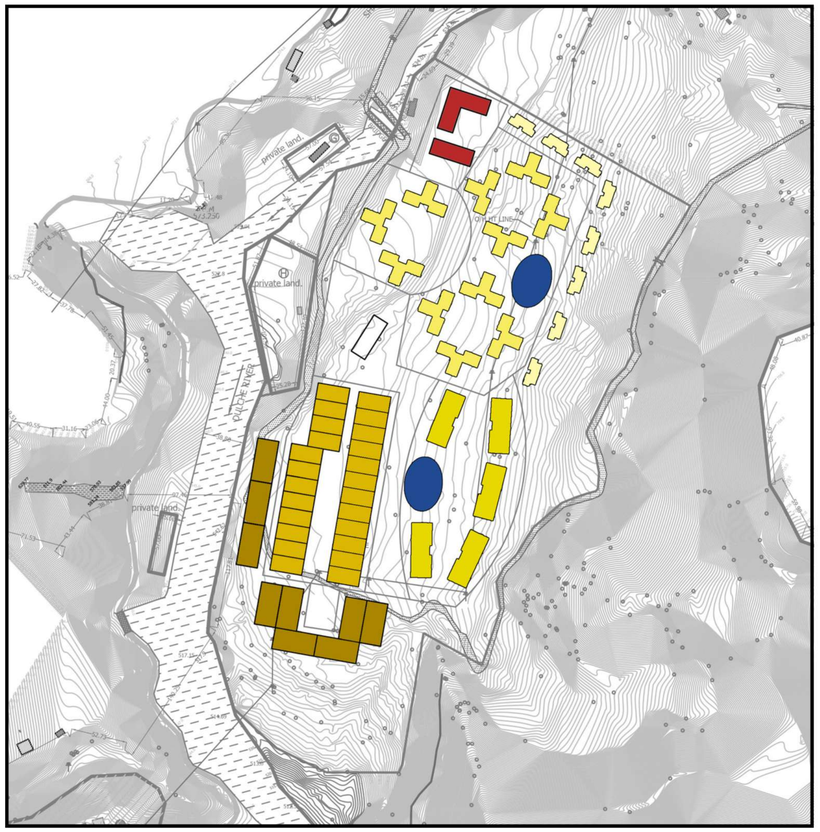


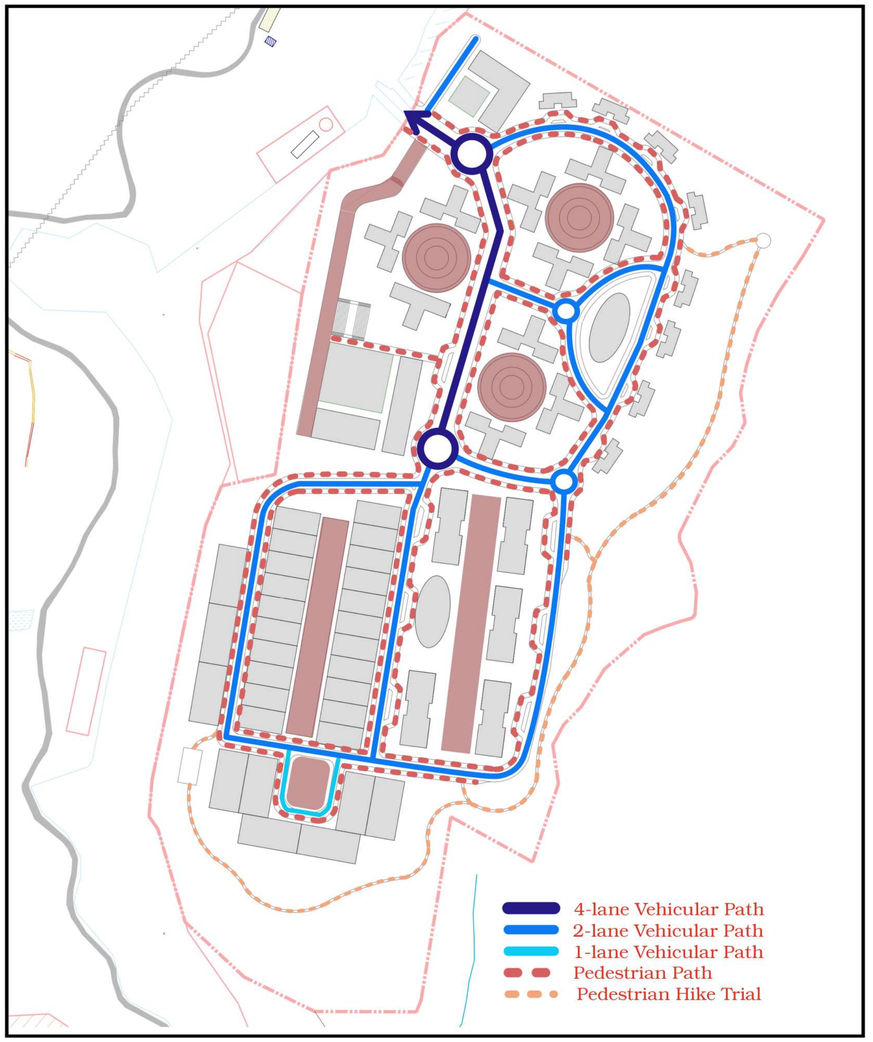
SLOPE ANALYSIS



ZONING & CIRCULATION LAYOUT







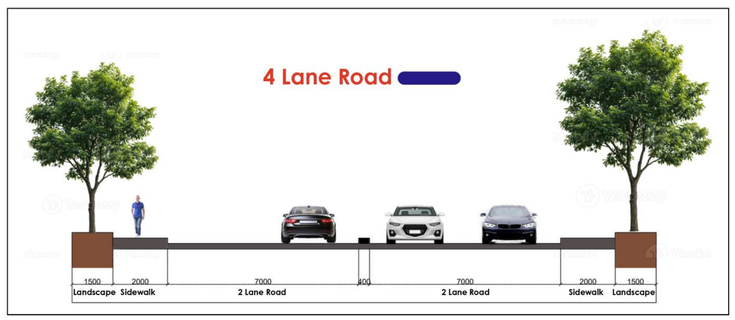
AREA PROGRAM

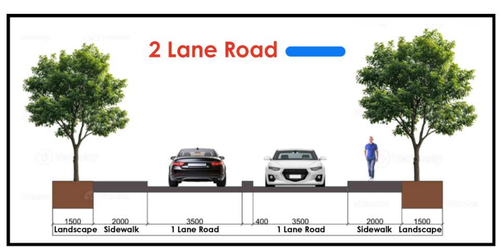
Type	Footprint	No.of Towers	No.of Floors
Basic	300	8	10
Medium	500	10	12
Premium	800	5	3(11), 2(5)

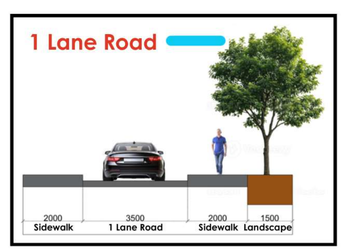
Type	Footprint	No.of Towers	No.of Floors
Commercial	2000	9	4
Amenities	2000	2	2

Type	Footprint
Plotted A	450
Plotted B	800

- COMMERCIAL
- BASIC BUILDING
- MEDIUM BUILDING
- PREMIUM BUILDING
- PLOTS
- AMENITIES







The steep and uneven topography of the site created several planning difficulties. Large level differences made it challenging to place buildings and roads without excessive cutting and filling of the land. Designing vehicular circulation was also difficult because road gradients had to remain safe and accessible while following the natural slope.
Additionally, low lying areas of the site were prone to water accumulation due to natural drainage patterns, which required careful placement of buildings and open spaces. These conditions required the zoning and circulation network to respond closely to the natural contours of the terrain.

MANDI LANDSCAPING

Site Risks:

- Landslides and Slope Failure
- Surface Soil Erosion
- Water Saturation and Runoff
- Wildfire Risk
- Extreme Temperatures

Site Risks	Plant List Solution & Specific Species	Mechanism of Action
Landslides & Slope Failure	Trees: Ban Oak (Q. leucotrichophora), Himalayan Cedar (C. deodara)	Deep Root Anchorage: The deep, woody root of these large trees anchor, binding the soil mass, thus increasing the shear strength and resistance to deep slope failure.
Surface Soil Erosion	Groundcovers: Dodonaea (D. viscosa), English Ivy (H. helix), Vetiver Grass	Surface Netting: These plants form a dense, fibrous layer on the topsoil. This mat dissipates raindrops and slows water runoff, preventing soil erosion.
Water Saturation & Runoff	Trees: Neem, Horse Chestnut, Jacaranda. Shrubs: Rhododendron, Crape Jasmine.	Hydrological Reinforcement: Tree canopies intercept rainfall. Their roots actively absorb water from the soil, helping reduce the overall soil saturation.
Wildfire Risk	Trees: Ban Oak (Quercus leucotrichophora), Neem, Dodonaea Viscosa	Fire Resistance: These trees have a naturally thick bark and high moisture content and dense leaves, highly resistant to catching fire, creating a firebreak.
Extreme Temperatures	Diversity & Native Adaptation: Lantana, Neem, Bougainvillea, Himalayan Cedar, Rhododendron, Horse Chestnut.	Resilience: These plants thrive in summers and winters, ensuring year-round green cover and reducing the risk of complete plant die-off mid temperature fluctuations
Urban Aesthetics	Trees: Jacaranda. Shrubs: Bougainvillea, Crape Jasmine.	Color & Fragrance: These ornamental species create an attractive, welcoming environment with spectacular seasonal colors and fragrances.

The landscape design was planned carefully in response to the site's ecological conditions, particularly its vulnerability to forest fires. Native trees and plant species were selected because they are naturally adapted to the region and have higher moisture content, making them more resistant to fire.

Local species were also chosen to preserve the natural character and beauty of the surrounding hills. Larger trees were strategically planted along roads and common areas to provide shade, improve comfort in public spaces, and strengthen the overall landscape identity of the site.

TREES


BAN OAK


JACARANDA


HIMALAYAN CEDAR


NEEM


HORSE CHESTNUT

SHRUBS


BOUGAINVILLEA


CRAPE JASMINE

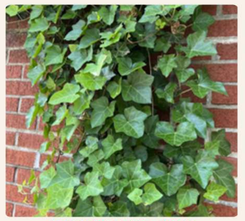

NIRGUNDI


LANTANA


RHODODENDRON

GROUNDCOVER


DODONAEA VISCOSA

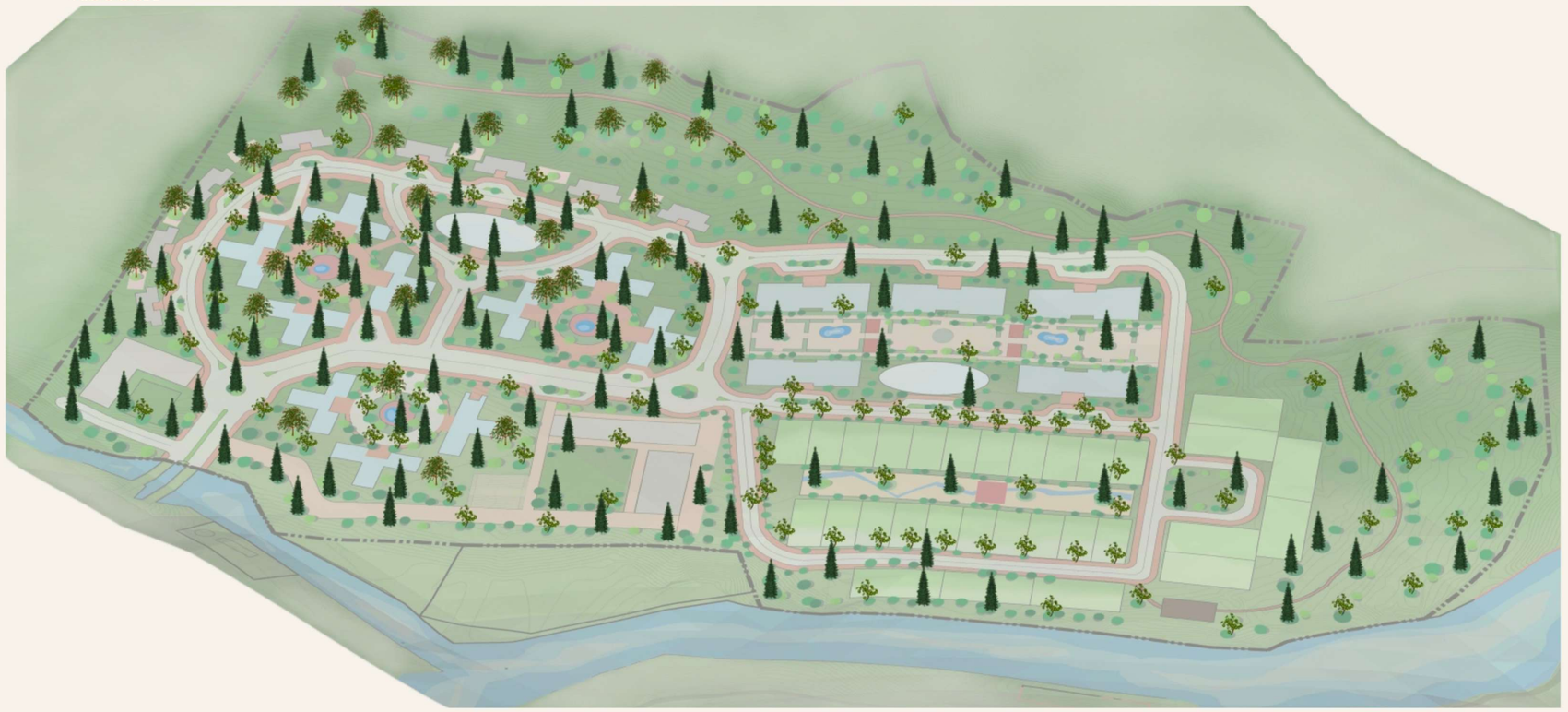

ENGLISH IVY


SEDUM

NATIVE GRASSES


VETIVER


ANDROPOGON



MANDI MASTER PLAN



Thankyou

